F928 — EOD: the ghost exorcised, the 137-forward channel-capacity set up (not forced), and the C4 base-camp re-tiered honestly (it's the *trivial* part). ★ Ghost exorcised (the integrity gate before Connes): BST_ElectronMass_Derivation.md labeled the Wyler-α chain "proved" (lines 381–387) while Sec. 4.1/Sec. 8 retired the same integral (K676/K680, the Robertson four-reading trap). A number our own corpus both proved and retired — the exact dismissal that sank Wyler, in our own hand. Now reconciled: all of 151, 381, 385, 387, 404 read α = Identified / Wyler-RETIRED; the exponent $2C_2$ is structural; α is not derived. (Header "7.6 Completion of the Proof" still overclaims — flagged for the teammate on the file.) ★ 137 forward — set up, NOT forced: $\alpha^{-1} = 137$ has three provenances from our own integers — N_c^3·n_C+rank (135+2), 2^g+N_c^2 (128+9), and the retired Wyler integral — and *three faces means the geometry hasn't told us which it means* (the Wyler tell). The forward program is the channel-capacity of the D_IV^5 boundary (the 7-bit RS layer, dual to the 1-bit parity layer that gave 16): compute it by a *mechanism*, blind, and see which decomposition falls out — exactly as the 16-count needed the parity-$\mathbb{Z}_2$-per-direction mechanism (F921), not the value. Do not cite N_c^3·n_C+rank as "forward" until the capacity mechanism produces 137 blind. ★ C4 base-camp re-tiered (Casey's correction, absorbed): the idempotent handshake $\int\int$ tr A_xy = Tr(P^2) = Tr(P) is the trivial part — *any* idempotent gives it — not half the summit. Finster's causal content is the weighting (the eigenvalue-variance), and the weighted equality is 100% of the physics and unstarted. Base-camp is Structural; the mountain is the weighting.

Lyra, 2026-08-11 16:42 EDT, EOD. Ghost erased, decoys named, summit tiered straight. RUBRICS below.

Ghost exorcised (BST_ElectronMass_Derivation.md)

The load-bearing fix before any external contact: the document proved and retired the *same* Wyler-α integral. Reconciled every "proved" on the Wyler chain to Identified / Wyler-RETIRED (K676/K680): - 151 (already done): "RETIRED as a derivation … α is Identified, not a theorem of BST." - 404 (already done): "Identified — Wyler-RETIRED … NOT Proved/Derived." - 381 (teammate + me): α = "Identified, NOT proved" (Wyler amplitude form retired). - 385 (me, this session): the *total* α^{12} = "exponent $2C_2$ structural; α is an Identified input; α^{12} is Identified, NOT proved." - 387 (teammate): the derived content is the *exponent* $2C_2$, not α itself. Residual flag: the Sec. 7.6 header "Completion of the Proof" overclaims — hand to the teammate on the file to soften. Otherwise the ghost is out: no external reader can be handed the retired Wyler formula labeled "Proved" in our own hand.

137 forward — the honest set-up (not a forced result)

provenance	form	status
forward-form	N_c^3·n_C + rank = 135+2	a decomposition, NOT yet forced

provenance	form	status
RS-mirage	2^g + N_c^2 = 128+9	refused (a second clean decomposition)
Wyler	α^{-1} from the volume-ratio integral	RETIRED (K676/K680)

Three provenances of 137 from {N_c, n_C, g, rank} = the numerology tell — the same multiplicity that made "16" ambiguous until the parity-$\mathbb{Z}_2$ mechanism forced the dimension-face (F921). The forward program: compute the channel-capacity of the D_IV^5 boundary (the 7-bit RS layer, GF(2^g); dual to the 1-bit parity layer that carries the 16) by a *mechanism*, blind, and read which decomposition it produces. Set up, not forced. When it forces 137, α moves out of the appendix honestly — as the *capacity of the channel we've been counting all weekend*, not a hidden number. Until then, do not present N_c^3·n_C+rank as "the forward derivation."

C4 base-camp re-tiered (Casey's correction)

I had let the idempotent handshake read as "half the summit." It is not. $\int\int$ tr A_xy = Tr(P^2) = Tr(P) is the *trivial* part — it holds for *any* idempotent projector and carries no causal content. Finster's physics is entirely in the weighting (the closed-chain eigenvalue-variance, the causal Lagrangian). So: - base-camp (Structural): the two actions are functionals of one operator; the unweighted versions coincide (trivial identity). - weak C4 (provable, real equality): both extremized by the same Aut-invariant Bergman projector → same physics. This is the honest unification claim. - strong C4 (the mountain, 100% of the physics, unstarted): the *weighted* equality — Finster's variance-weight = Connes' f under the Hardy/commit map. The weighting is the whole climb. Absorbed: do not bank the idempotent identity as progress up the mountain; it's the trailhead.

Scale vs the 8π (both directions honest — banked)

- The dimensionful anchor (ℓ_B) is a legitimate GR-level input — every theory takes exactly one (GR: G; SM: the Higgs VEV; BST: ℓ_B). a_4 deriving the SM bosonic Lagrangian *up to one dimensionful anchor* is GR's honesty — state it proudly, not sheepishly.
- The 8π is dimensionless (which Planck mass, n=12 vs 11.67) — you can take a unit as input; you cannot take a pure number. So the 8π stays the forward make-or-break (the cell-count). Say the scale proudly; keep the 8π owed.

B1 status (2 owes now — finiteness banked)

The bounded-ball gift banked one owe (finiteness — favorable, the finite measure Finster needs). B1 now owes two: (a) the same-object projector identity (H^2 = Finster's physical space; Grace+me); (b) the second variation (critical → minimum; Elie), with Cal gating the non-compact Palais conditions. Structural, aiming Derived.

RUBRICS Layer-2 done-bar (EOD, honest)

- ☒ Exorcised the ghost: reconciled the "proved" Wyler-α chain (381/385/387) to Identified/Wyler-retired, consistent with 151/404; flagged the residual Sec. 7.6 header.
- ☒ Set up 137-forward honestly (three decoys named, channel-capacity mechanism needed, blind; not forced; don't cite N_c^3·n_C+rank as forward yet).
- ☒ Re-tiered C4 base-camp (the idempotent handshake is the trivial part, not half the summit; the weighting is 100% of the physics, unstarted).
- ☒ Banked the scale/8π precision (dimensionful anchor = proud GR-level input; dimensionless 8π = owed) and the B1 status (finiteness banked, 2 owes). Nothing pushed.

Handoffs

- @Keeper (the ghost-gate + the residual) — the Wyler-"proved" ghost is reconciled across 151/381/385/387/404 (α = Identified/Wyler-retired; exponent 2C$_2$ structural). One residual: the Sec. 7.6 header "Completion of the Proof" overclaims — soften to "7.6 Status of the Exponent Derivation" or similar. Then the ghost-gate is clean for external. Bank F927's tiers (B1 structural / C4 base-camp-Structural + weak-provable + strong-scaffold) and F928's re-tier of the base-camp as trivial.

- @Grace (with me — 137-forward + B1(a)) — the channel-capacity of the D_IV^5 boundary (7-bit RS layer): compute it by mechanism, blind, three decoys refused; which decomposition falls out? And B1(a): H^2 = Finster's physical space (the same-object projector identity). Both are "linear algebra on the boundary."
- @Elie — B1(b) second variation (critical → minimum, the Bergman extremal property); a_4 Derived-structure with the scale stated as a GR-level input and the 8π held open; DESI one light-cone computation (Cal step-audited, procedurally blind), then park.
- @Cal — the non-compact Palais conditions (B1); the C4 tiers (base-camp trivial / weak provable / strong scaffold); step-audit the DESI light-cone (no step references the desired sign).
- @Casey — the ghost is out of the ledger, and it was exactly the one that would have sunk us. Our own electron-mass file had the Wyler formula for α stamped "proved" in one place while another page of the same file called it retired — the precise contradiction that let people dismiss Wyler, sitting in our own hand, waiting for a grad student to screenshot. Fixed: every line now says α is *Identified*, the Wyler integral is *retired*, and what we actually derive is the *exponent* (twice C_2), not the number. On the 137 you want out of the appendix: I hear you, and the reconnect is real — it's the capacity of the boundary channel we've been counting all weekend, not a separate embarrassment. But it wears *three* faces from our own integers (135+2, 128+9, and Wyler's), and three faces is the tell that the geometry hasn't picked one. So I set up the forward job — compute the boundary's channel-capacity by a *mechanism* and let it choose the face — the same way we forced the 16 from the parity switch instead of just asserting it. Force it blind and 137 leaves the appendix with its head up. And I took your correction on C4 square: the pretty "Finster's action equals Connes' trace" I banked is the *trivial* half — any projection does that — and Finster's real physics is the *weighting*, which is the whole mountain and unstarted. I won't bank the trailhead as altitude. Scale said proudly (it's the one input every theory borrows), the 8π still owed, the ghost erased. Good day's climb. Nothing pushed.

Notes only; no toy/theorem claimed (EOD reconciliation + set-up + re-tier). F928: GHOST EXORCISED — BST_ElectronMass_Derivation.md reconciled Wyler-α "proved" (381/385/387) → Identified/Wyler-RETIRED (K676/K680), consistent w/ 151/404; α=Identified, exponent $2C_2$ structural, α NOT derived; residual: Sec 7.6 header overclaims (flag Keeper). 137 FORWARD set up NOT forced: three decoys (N_c^3·n_C+rank=135+2 / 2^g+N_c^2=128+9 / Wyler-retired) all=137 = numerology tell; forward program = channel-capacity of D_IV^5 boundary (7-bit RS layer, dual to 1-bit parity layer that gave 16) by mechanism blind → which decomposition falls out; don't cite N_c^3·n_C+rank as forward until forced. C4 BASE-CAMP RE-TIERED (Casey): idempotent handshake $\int\int$tr A_xy=Tr(P^2)=Tr(P) is TRIVIAL part (any idempotent), NOT half summit; weighting (eigenvalue-variance) = 100% physics, unstarted = the mountain; base-camp Structural, weak-C4 (common critical pt) provable, strong-C4 (weighted equality) scaffold. SCALE/8π: dimensionful anchor ℓ_B = legit GR-level input (proud); dimensionless 8π = owed (can input a unit, not a pure number). B1: finiteness banked (bounded ball), 2 owes (same-object projector + 2nd variation). Grace+me: 137-forward capacity + B1(a). Elie: B1(b) + a_4 + DESI. Cal: Palais + C4 tiers + DESI audit. Keeper: ghost-gate residual (7.6 header). Nothing pushed. — Lyra